Boulder County

# Land Use

Courthouse Annex • 2045 13th Street • Boulder, Colorado 80302 • Tel: 303.441.3930 • Fax: 303.441.4856
**Mailing Address:** P.O. Box 471 • Boulder, Colorado 80306 • **www.bouldercounty.org**

February 26, 2015

Patricia Thornton
Project Manager
Powder River Development
219 S Wooddale Ave
Eagle ID 83616

Re: T-MOBILE DN03035D
Minor Modification Request for SU-06-016 at the existing facility located at 5134 Ute Hwy parcel id 120328000001

Dear Patricia Thornton,

Staff has reviewed the materials provided has found that the proposal is an eligible request for modification per 2014 FCC rules pertaining to Section 6409(a) of the Middle Class Tax Relief and Job Creation Act of 2012 . The proposal constitutes an eligible facilities request because the existing Roof Top Facility is proposed for collocation of new transmission equipment, and/or removal of transmission equipment, and/or replacement of transmission equipment.

Additionally, staff has determined that the proposal is a non-substantial change as defined by the 2014 FCC rules pertaining to Section 6409(a) of the Middle Class Tax Relief and Job Creation Act of 2012.

Because local governments must approve any eligible facilities request for a modification that does not substantially change the physical dimensions of the existing facility, this proposal is approved.

Per the 2014 FCC rules modifications must maintain concealment elements and comply with conditions associated with the prior approval of the facility. In order to meet this requirement the following conditions shall apply to this approval:

1. A 7 year review hearing is required by the Special Use Review docket approval SU-06-016. Provided an appointment is made to schedule the 7 year review followed by a submittal for review, building permits will not be withheld. Please contact our Receptionist at 303-441-3930 to schedule a pre-application appointment with a Current Planner for a 7 year Special Use Review.

2. All necessary building permits for the structure and components shall be obtained and the structures and components shall comply with Boulder County Building Codes.

2. The antennas shall not exceed 167' 0' (163' 0" to center line of the antenna).

3. All additional ground equipment will be housed within the existing equipment shelter.

**Cindy Domenico** *County Commissioner* **Deb Gardner** *County Commissioner* **Elise Jones** *County Commissioner*

4. The colors for all proposed components shall match the existing tower and equipment and be equal in hue, value, and chroma.

5. All equipment, hardware, etc., that is no longer in use by T-Mobile on the site shall be removed from the site. If I may be of further assistance, please call me at 303-441-3930.

Regards,

Bonnie Gracia

Bonnie Gracia
Planner On-call

February 9, 2015

Boulder County
Planning and Development Services Center
2045 13th St
Boulder, CO 80302

RE: Administrative Review Application for Antenna for Wireless Facility Written Statement
5134 Ute Highway, Lyons CO 80503

To Whom It May Concern:
T-Mobile is proposing to modify its existing rooftop wireless facility at 5134 Ute Highway in Lyons.
Parcel Number 120328000001

The proposed modifications will include the removal of the six (6) antennas.
Install of nine (9) new antennas, and nine (9) new remote radio units are to be added.

Inside the equipment shelter install 4 (4) COVP's and one (1) stacked sleeve are to be added.

Neither the height nor footprint of the existing site will change with the proposed scope of work. Access to the site, power, and equipment shelter already exist at the site.

Please find enclosed for your review:

- one (1) copy of Zoning Drawings (11" x 17")
- one (1) copy Written Statement

Please feel free to contact me at any time with any questions or concerns that you may have.
I look forward to working with you on this project.

Best Regards,

Patty Thornton

Project Manager

801-860-1219 Cell

patricia.thornton@powderriverdev.com

*Enclosures*

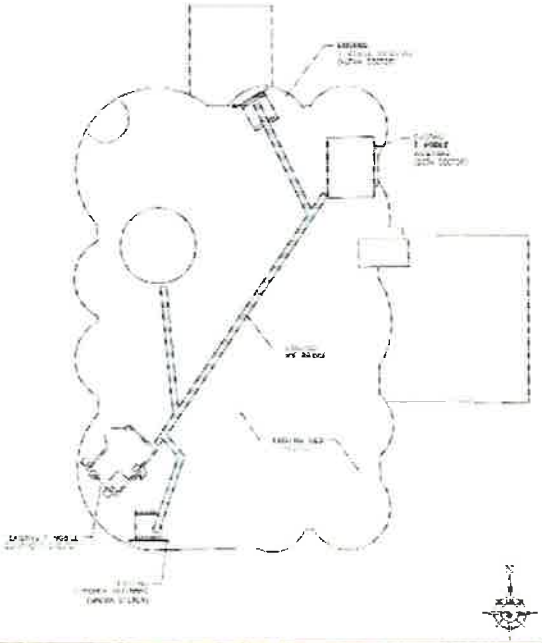
# 700 MHz Equipment Upgrade

T-Mobile

POWDER RIVER
Development Services, LLC

**DN3035D**
SOUTHDOWN LYONS
5134 UTE Highway
Lyons, CO 80540







# 700 MHz Equipment Upgrade

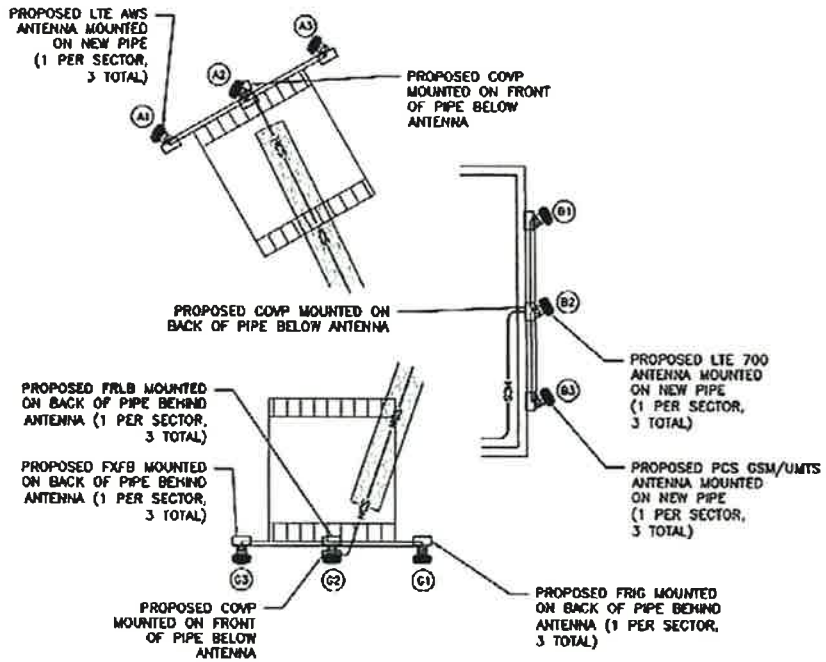
T-Mobile

POWDER RIVER
Development Services, LLC

**DN3035D**
SOUTHDOWN LYONS
5134 UTE Highway
Lyons, CO 80540



Existing Elevation




Plan View



Proposed Elevation

DN03035D

T-Mobile

**SENDER: COMPLETE THIS SECTION**

- Complete items 1, 2, and 3. Also complete item 4 if Restricted Delivery is desired.
- Print your name and address on the reverse so that we can return the card to you.
- Attach this card to the back of the mailpiece, or on the front if space permits.

1. Article Addressed to:

Southdown, Inc
attn: Plant Manager
5134 Ute Highway
~~Lyons~~ Longmont CO 80503
NOTICE DN03035D

**COMPLETE THIS SECTION ON DELIVERY**

A. Signature
X Marianne Smith ☐ Agent ☐ Addressee

B. Received by (Printed Name) Marianne Smith C. Date of Delivery 9-10-14

D. Is delivery address different from item 1? ☐ Yes
If YES, enter delivery address below: ☐ No

3. Service Type
☒ Certified Mail ☐ Express Mail
☐ Registered ☐ Return Receipt for Merchandise
☐ Insured Mail ☐ C.O.D.

4. Restricted Delivery? (Extra Fee) ☐ Yes

2. Article Number (Transfer from service label) 7011 0470 0003 6939 5939

PS Form 3811, February 2004 Domestic Return Receipt 102595-02-M-1540

**VIA CERTIFIED U.S. MAIL**
**RETURN RECEIPT REQUE**

August 28, 2014

Southdown, Inc.
Attn: Plant Manager
5134 Ute Highway
Lyons, CO 80503

**Re: Site Lease with option dated March 9, 2000 between Southdown, Inc. and T-Mobile West Corporation, successor in interest to VoiceStream PCS II Corporation.**

T-Mobile Site: DN03035D – Southdown Lyons
Site location: 5134 Ute Highway, Lyons, CO 80503

Dear Sir or Madam:

The Site Lease Agreement authorizes T-Mobile to use its premises for the operation and maintenance of its wireless communications facility. We value our landlords and are providing you with this letter regarding our upcoming maintenance as a courtesy.

While the work entails the replacement of existing items of equipment and the addition of new items, the work is not very different from routine maintenance work that we do, and will not interfere with your business or tenants. The work will be performed in compliance with the non-interference requirements of our lease and with all applicable laws and regulations. Under the lease, your approval is not required and you do not need to take any further action regarding this letter.

If you have any questions, you may call Tom Walker at 303-868-6866 or email Tom.Walker@powderriverdev.com.

Sincerely,

Scott W. DuBuke
Director, Network Engineering & Operations
T-Mobile – Denver Market

T··Mobile·

# SITE NAME: SOUTHDOWN LYONS
# SITE NUMBER: DN03035D

## EQUIPMENT UPGRADE

### SITE INFORMATION

| | |
|---|---|
| PROPERTY OWNER: | CEMEX INC. |
| TOWER OWNER: | CEMEX INC. |
| SITE NAME:<br>SITE NUMBER: | SOUTHDOWN LYONS<br>DN03035D |
| SITE ADDRESS: | 5134 UTE HIGHWAY<br>LYONS, CO 80540 |
| COUNTY: | BOULDER |
| LATITUDE:<br>LONGITUDE:<br>LAT/LONG TYPE: | 40.203586<br>−105.235792<br>NAD 83 |
| GROUND ELEVATION: | 5222 AMSL |
| ZONING JURISDICTION: | BOULDER COUNTY |
| PARCEL ID: | 120328000001 |

### CONTACT INFORMATION

ENGINEERING:
POWDER RIVER DEVELOPMENT SERVICES, LLC.
219 S. WOODDALE AVE.
EAGLE, ID 83616
PHONE: 303.410.6616
CONTACT: TOM FULTON

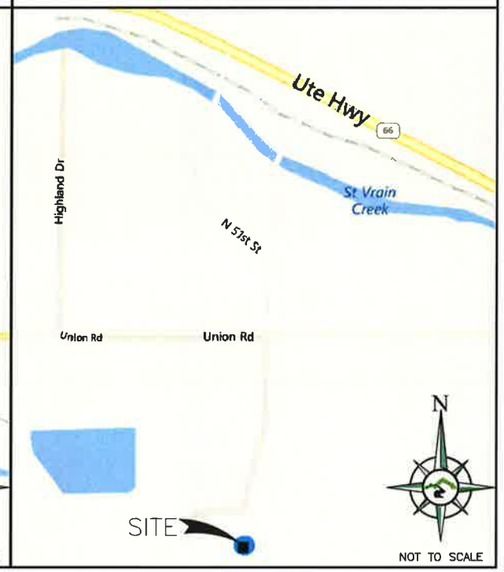
APPLICANT:
T−MOBILE
2323 DELGANY STREET
DENVER, CO 80216
PHONE: 303.313.6930

SITE ACQUISITION:
POWDER RIVER DEVELOPMENT SERVICES, LLC.
219 S. WOODDALE AVE.
EAGLE, ID 83616
PHONE: 303.868.6866
CONTACT: TOM WALKER

### VICINITY MAP




### LOCAL MAP




### DRIVING DIRECTIONS

FROM DENVER INTERNATIONAL AIRPORT:

DEPART TOWARD PENA BLVD.
KEEP STRAIGHT ON TO PENA BLVD.
AT EXIT 6B, TAKE RAMP RIGHT FOR E−470 TOLLWAY NORTH TOWARD BOULDER/FT.COLLINS
AT EXIT 47, TAKE RAMP RIGHT FOR I−25 NORTH TOWARD FT COLLINS
AT EXIT 243, TAKE RAMP RIGHT FOR CO−66 TOWARD ROCKY MOUNTAIN NATIONAL PARK/PLATTEVILLE/LYONS
TURN RIGHT ONTO CO−66W
KEEP STRAIGHT ONTO CO−66
KEEP STRAIGHT ONTO CO−66/UTE HWY
TURN LEFT ONTO N 51ST ST
KEEP STRAIGHT ONTO ROAD
ARRIVE AT 40.203586 −105.235792 ON THE LEFT

### DRAWING INDEX

| SHEET #: | DESCRIPTION: |
|---|---|
| T−1 | TITLE SHEET |
| A−1 | SITE PLAN AND EQUIPMENT PLAN |
| A−2 | ELEVATIONS |
| A−3 | ELEVATIONS |
| A−4 | ANTENNA PLANS |
| A−5 | ANTENNA DETAILS |
| RF−1 | RF CONFIGURATION |
| G−1 | GROUNDING DETAILS |
| G−2 | GROUNDING DETAILS |
| S−1 | STRUCTURAL DETAILS |

### APPLICABLE CODES

BUILDING CODE − 2012 IBC
ELECTRIC CODE − 2014 NEC

### DO NOT SCALE DRAWINGS

CONTRACTOR SHALL VERIFY ALL PLANS & EXISTING DIMENSIONS & CONDITIONS ON THE JOB SITE & SHALL IMMEDIATELY NOTIFY THE ENGINEER IN WRITING OF ANY DISCREPANCIES BEFORE PROCEEDING WITH THE WORK OR BE RESPONSIBLE FOR SAME

### PROJECT DESCRIPTION

MODIFICATION TO EXISTING NON−INHABITABLE TELECOMMUNICATIONS SITE CONSISTING OF THE REMOVAL OF (6) ANTENNAS, THE ADDITION OF (9) ANTENNAS, (9) RADIO UNITS, (4) COVPS, (3) HCS CABLES AND ASSOCIATED SYSTEM MODULES.

COLORADO 811

T··Mobile·

POWDER RIVER
Development Services, LLC
www.powderriverdev.com

LICENSE #: N/A
DRAWN BY: JMB

REVISIONS

| REV | DATE | DESCRIPTION | INT |
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| 1 | 02/06/15 | REVISED PER COMMENTS | JHP |
| 0 | 10/17/14 | 100% CONSTRUCTION | JMB |
| B | 10/15/14 | ISSUED FOR REVIEW | JMB |
| A | 09/30/14 | ISSUED FOR REVIEW | JMB |

THESE PLANS AND SPECIFICATIONS, AS INSTRUMENTS OF SERVICE, ARE AND SHALL REMAIN THE PROPERTY OF POWDER RIVER DEVELOPMENT SERVICES LLC WHETHER THE PROJECTS FOR WHICH THEY ARE MADE ARE EXECUTED OR NOT. THESE DRAWINGS AND SPECIFICATIONS SHALL NOT BE USED BY ANY PERSON OR ENTITY ON OTHER PROJECTS WITHOUT PRIOR WRITTEN CONSENT OF THE ENGINEER

SITE INFORMATION

**SOUTHDOWN LYONS**
**DN03035D**
**5134 UTE HIGHWAY**
**LYONS, CO**
**80540**

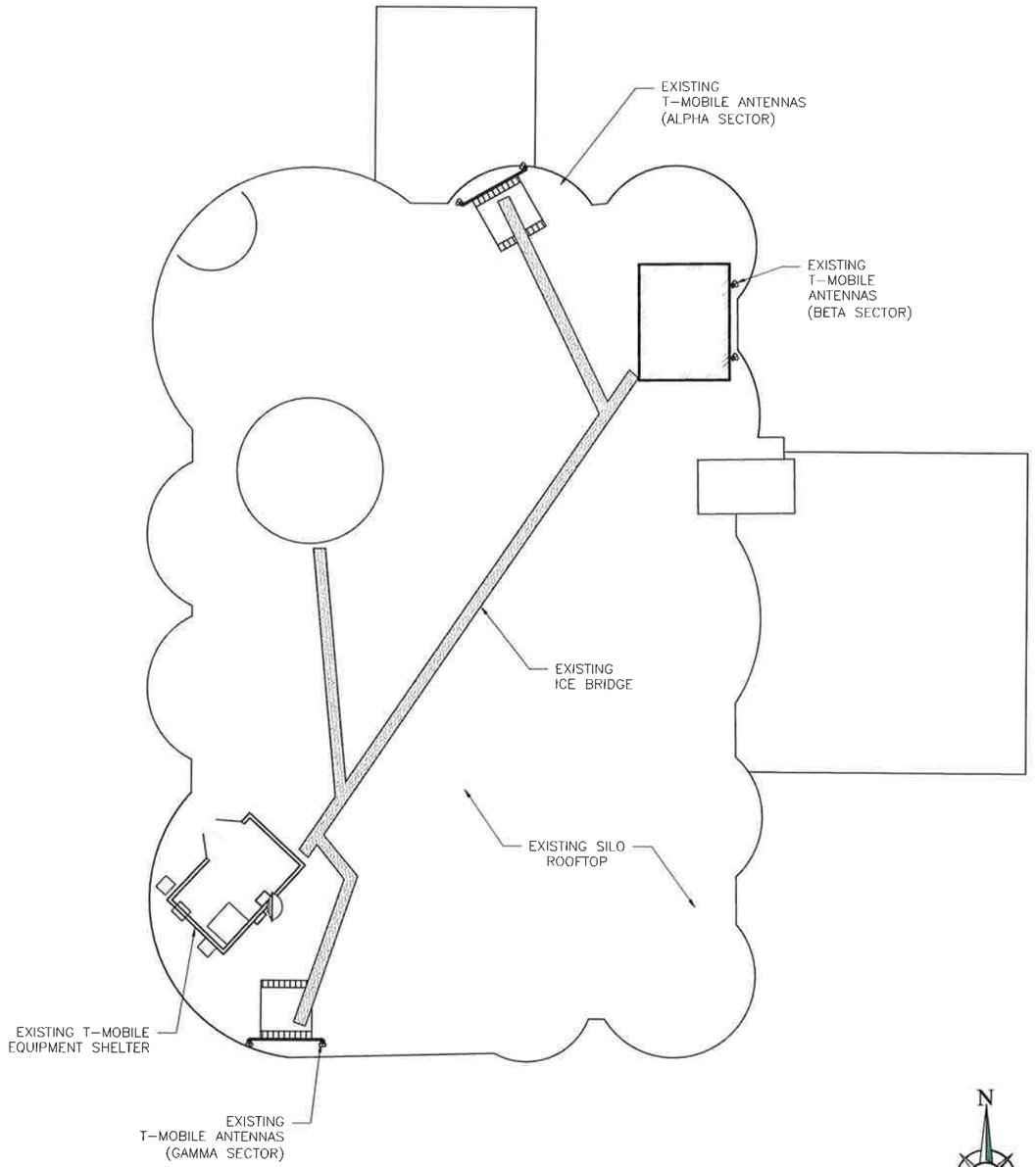
**EQUIPMENT UPGRADE**

SHEET TITLE:

TITLE SHEET

SHEET NUMBER:

T-1

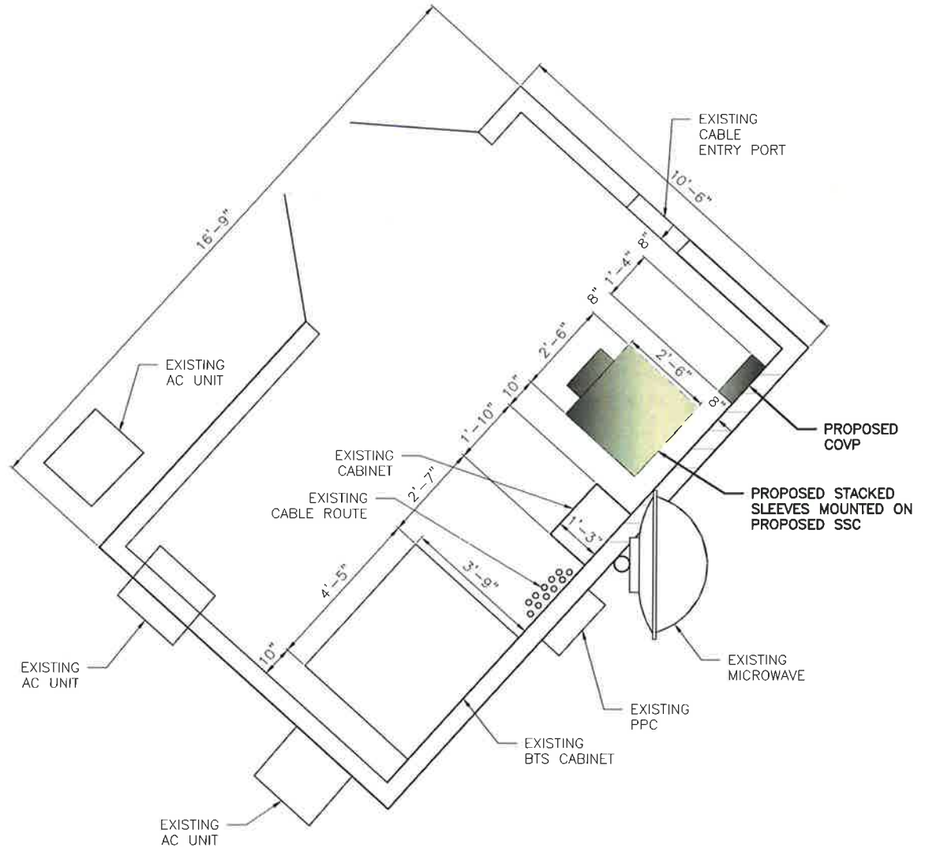



SITE PLAN (EXISTING)

10' 0 5' 10'

SCALE: 1" = 10' (24x36)
(OR) 1" = 20' (11x17)

2




EQUIPMENT PLAN (PROPOSED)

2' 0 1' 2'

SCALE: 1/2" = 1'-0" (24x36)
(OR) 1/4" = 1'-0" (11x17)

1

T-Mobile

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| LICENSE #: | N/A |
| DRAWN BY: | JMB |

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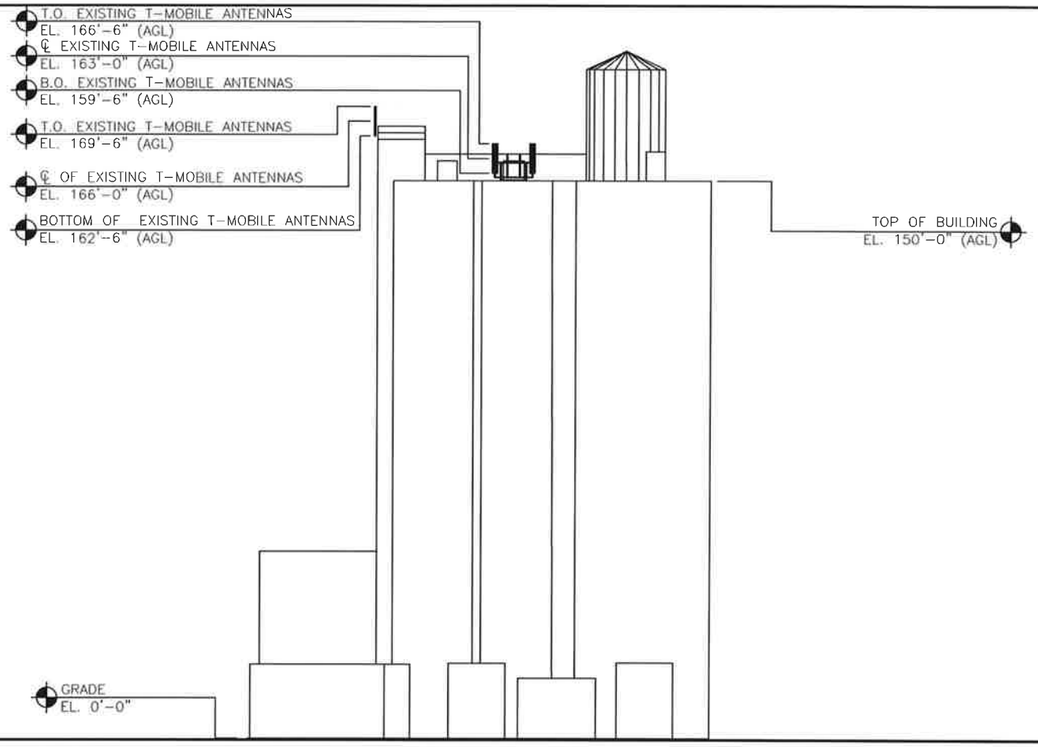
**EQUIPMENT UPGRADE**

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SITE PLAN
EQUIPMENT PLAN

SHEET NUMBER:

A-1



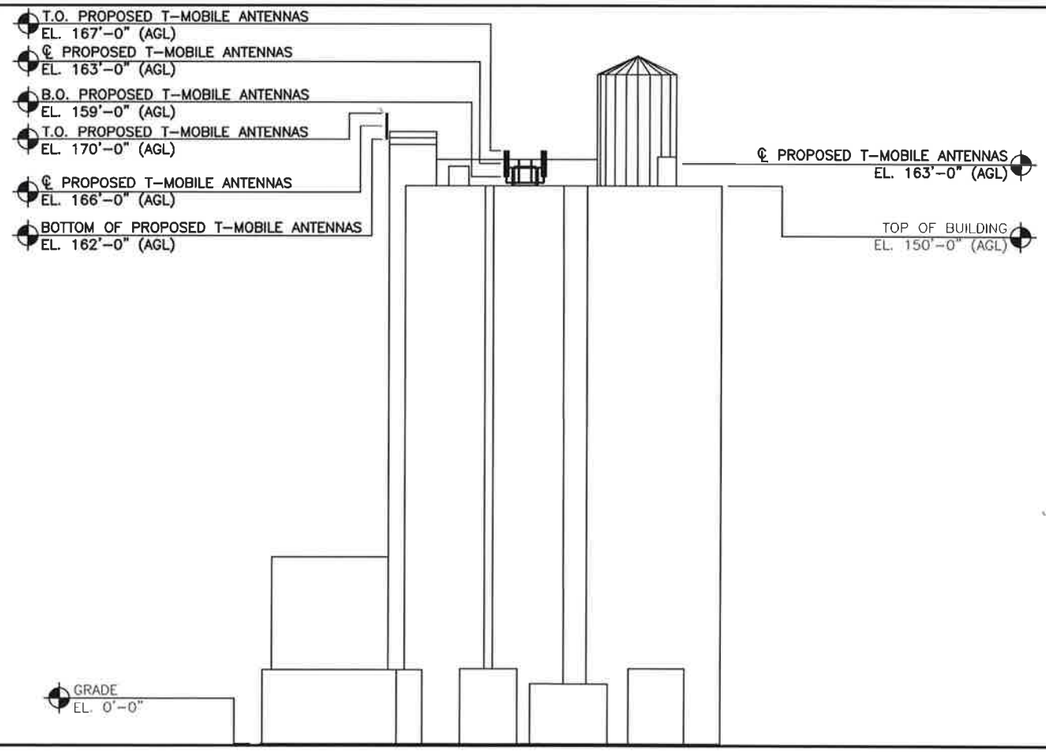

NOTE: STRUCTURAL ANALYSIS MUST BE PERFORMED ON ALL ROOFTOPS, FLAGPOLES, LIGHT POLES, AND TOWER SITES BEFORE INSTALLATION OF NEW ANTENNAS, NEW RRH'S, ETC.

NORTH ELEVATION (EXISTING)

20' 0 10' 20'

SCALE: 1" = 20' (24x36)
(OR) 1" = 40' (11x17)

1




NOTE: STRUCTURAL ANALYSIS MUST BE PERFORMED ON ALL ROOFTOPS, FLAGPOLES, LIGHT POLES, AND TOWER SITES BEFORE INSTALLATION OF NEW ANTENNAS, NEW RRH'S, ETC.

NORTH ELEVATION (PROPOSED)

20' 0 10' 20'

SCALE: 1" = 20' (24x36)
(OR) 1" = 40' (11x17)

2

T-Mobile

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www.powderriverdev.com

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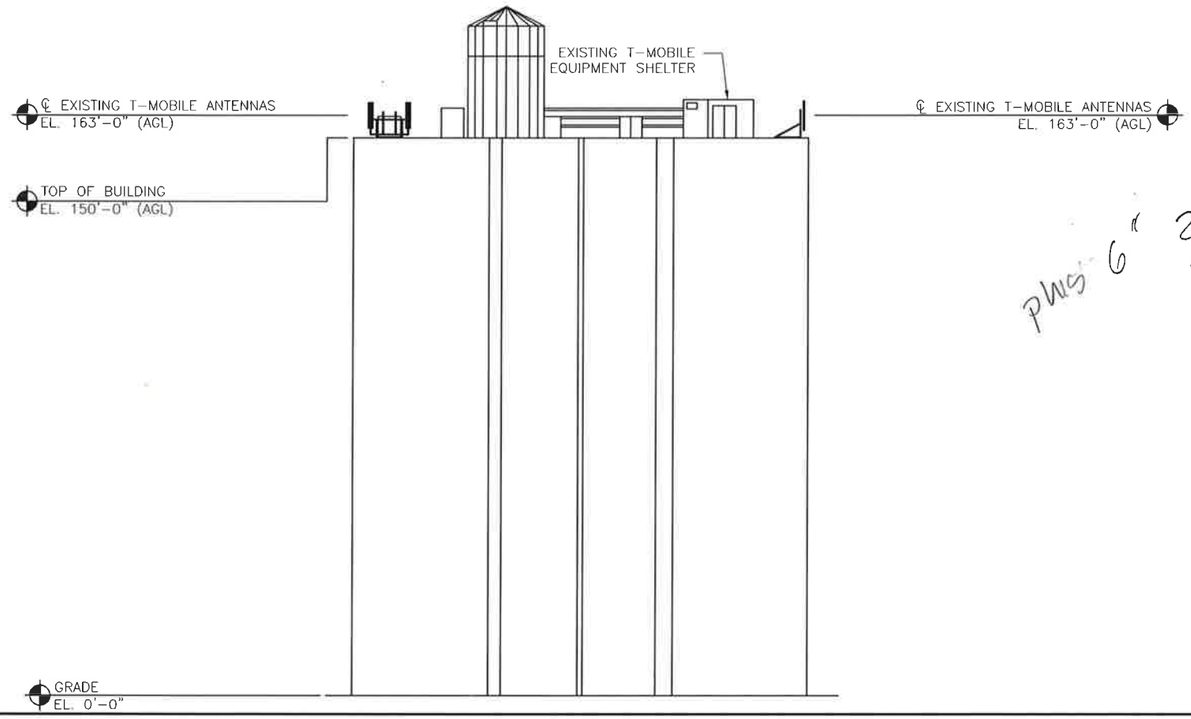
**EQUIPMENT UPGRADE**

SHEET TITLE:

ELEVATIONS

SHEET NUMBER:

A-2




NOTE: STRUCTURAL ANALYSIS MUST BE PERFORMED ON ALL ROOFTOPS, FLAGPOLES, LIGHT POLES, AND TOWER SITES BEFORE INSTALLATION OF NEW ANTENNAS, NEW RRH'S, ETC.

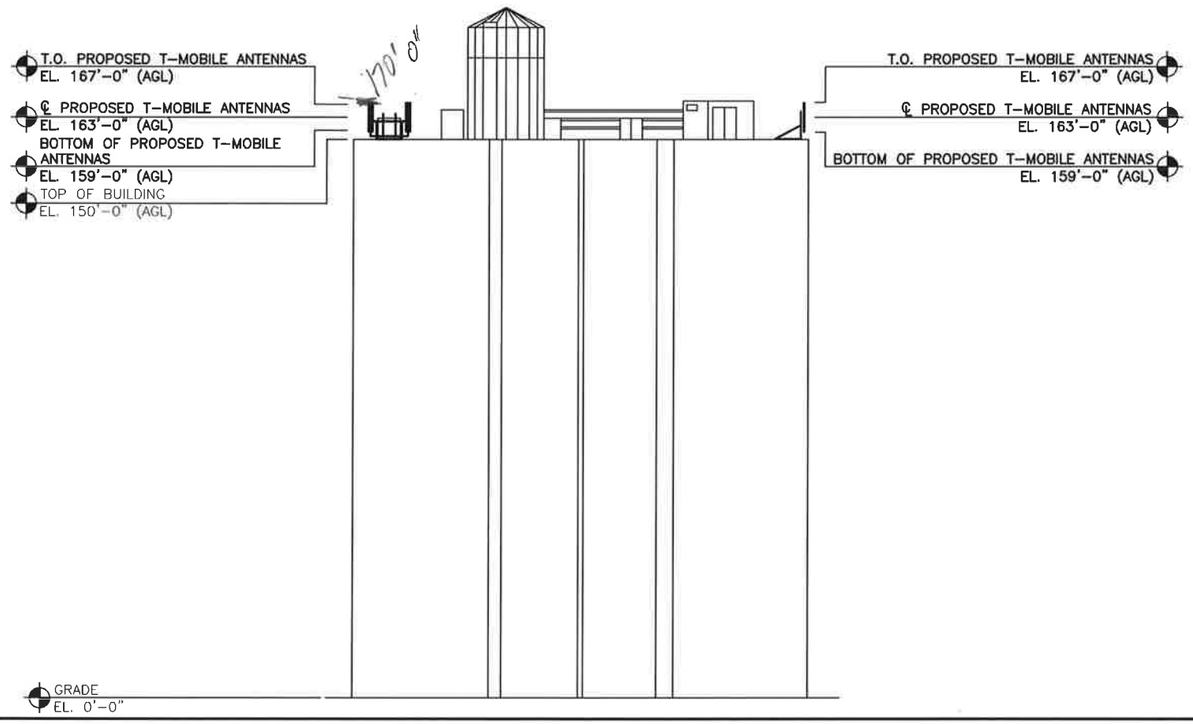
plus 6' 2 to 10

WEST ELEVATION EXISTING)

20' 0 10' 20'

SCALE: 1" = 20' (24x36)
(OR) 1" = 40' (11x17)

1




WEST ELEVATION (PROPOSED)

20' 0 10' 20'

SCALE: 1" = 20' (24x36)
(OR) 1" = 40' (11x17)

1

T-Mobile

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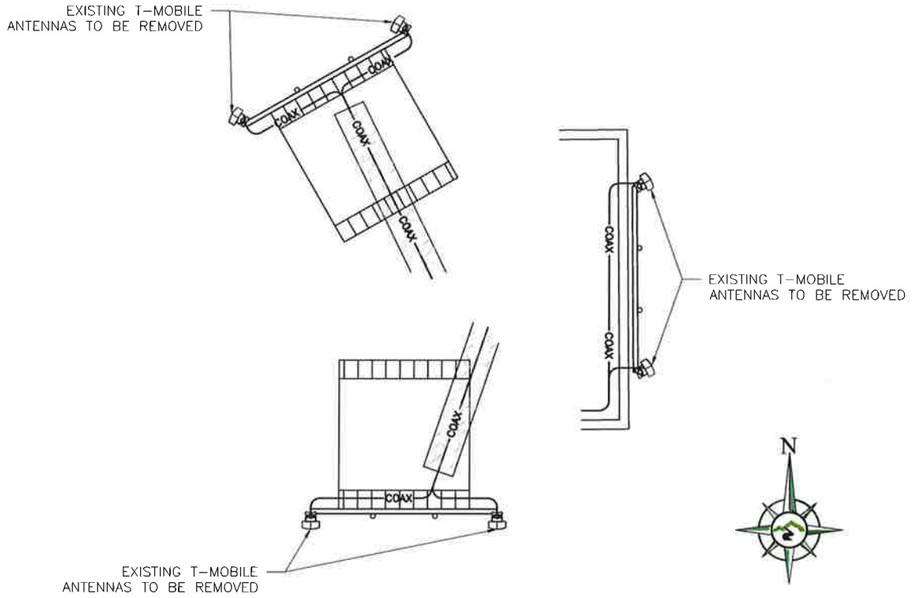
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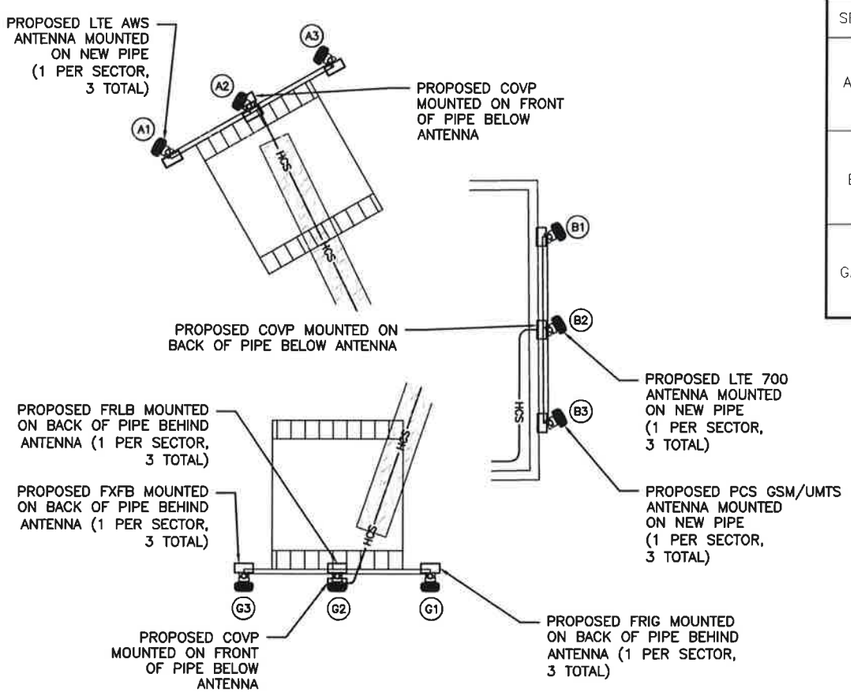
SHEET NUMBER:

A-3




## ANTENNA PLAN (EXISTING)

SCALE N.T.S. 1




ANTENNA EQUIPMENT SCHEDULE

| SECTOR | ANTENNA MARK | ANTENNA MODEL | RAD CENTER | AZIMUTH | TECHNOLOGY | RADIO MODEL | ANTENNA CABLE | CABLE LENGTH | HYBRID CABLE |
|---|---|---|---|---|---|---|---|---|---|
| ALPHA | A1 | ANDREW-TMBXX-6517-A2M | 156'-0" | 300° | LTE AWS | FRIG | LOW CAPACITY HCS - 1.24" | 50'-0" | FROM BETA |
| | A2 | ANDREW-LNX-6515DS-A1M | 156'-0" | 300° | LTE 700 | FRLB | | | |
| | A3 | ANDREW-TMBXX-6517-A2M | 156'-0" | 300° | PCS GSM/UMTS | FXFB | | | |
| | | | | | | | | | |
| BETA | B1 | ANDREW-TMBXX-6517-A2M | 166'-0" | 60° | LTE AWS | FRIG | LOW CAPACITY HCS - 1.24" | 125'-0" | FROM CENTRAL |
| | B2 | ANDREW-LNX-6515DS-A1M | 166'-0" | 60° | LTE 700 | FRLB | | | |
| | B3 | ANDREW-TMBXX-6517-A2M | 166'-0" | 60° | PCS GSM/UMTS | FXFB | | | |
| | | | | | | | | | |
| GAMMA | G1 | ANDREW-TMBXX-6517-A2M | 156'-0" | 180° | LTE AWS | FRIG | LOW CAPACITY HCS - 1.24" | 50'-0" | FROM CENTRAL |
| | G2 | ANDREW-LNX-6515DS-A1M | 156'-0" | 180° | LTE 700 | FRLB | | | |
| | G3 | ANDREW-TMBXX-6517-A2M | 156'-0" | 180° | PCS GSM/UMTS | FXFB | | | |
| | | | | | | | | | |

NOTE TO CONTRACTOR:

1. ANTENNA CLEARANCE AND MOUNTING TO BE FIELD VERIFIED PRIOR TO CONSTRUCTION WITH FINAL ANTENNA SPECIFICATIONS MOUNTING HARDWARD AND RF DESIGN. ANTENNA PIPE MOUNT MODIFICATION MAY BE REQUIRED.
2. CONTRACTOR TO VERIFY FINAL ANTENNA CONFIGURATION FROM FINAL RFDS.

## ANTENNA PLAN (PROPOSED)

SCALE N.T.S. 2

T-Mobile

POWDER RIVER
Development Services, LLC
www.powderriverdev.com

LICENSE #: N/A

DRAWN BY: JMB

REVISIONS

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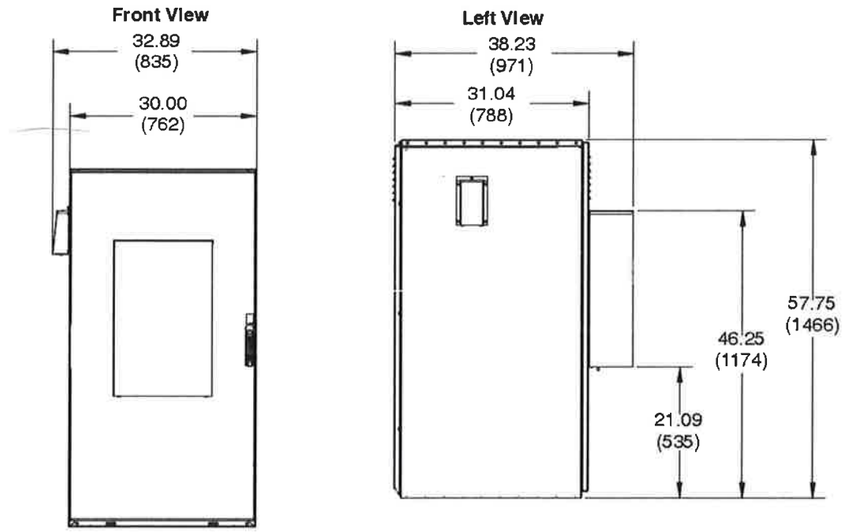
SITE INFORMATION

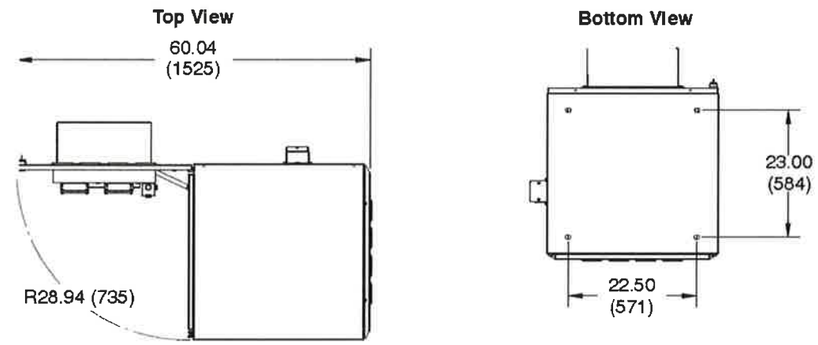
**SOUTHDOWN LYONS**
**DN03035D**
**5134 UTE HIGHWAY**
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**80540**

**EQUIPMENT UPGRADE**

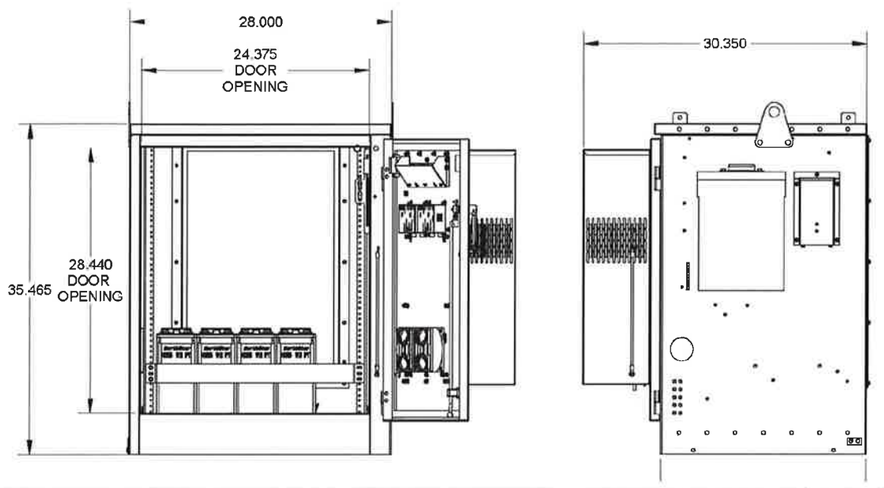
SHEET TITLE:
ANTENNA PLANS

SHEET NUMBER:
A-4

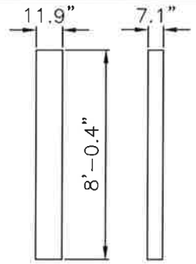



**MEDIUM SSC SFX31** | SCALE N.T.S. | 7




**SMALL SSC RA353W** | SCALE N.T.S. | 8



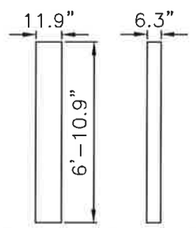

| ANDREW LNX-6515DS-A1M PROPOSED ANTENNA | |
|---|---|
| DEPTH | 7.1 IN |
| LENGTH | 96.42 IN |
| WIDTH | 11.9 IN |
| NET WEIGHT | 50.27 LB |
| COLOR | LIGHT GRAY |

**ANDREW LNX-6515DS-A1M** | SCALE N.T.S. | 4



| RRH 2x40W 730 MHz FXFB | |
|---|---|
| DEPTH | 5.25 IN |
| LENGTH | 22.0 IN |
| WIDTH | 19.3 IN |
| NET WEIGHT | 55.12 LBS |

**FXFB SPECIFICATIONS** | SCALE N.T.S. | 5




| ANDREW TMBXX-6517-R2M EXISTING ANTENNA | |
|---|---|
| DEPTH | 6.3 IN |
| LENGTH | 82.9 IN |
| WIDTH | 11.9 IN |
| NET WEIGHT | 44.4 LB |
| COLOR | OFF WHITE |

**ANDREW TMBXX-6517-R2M** | SCALE N.T.S. | 6



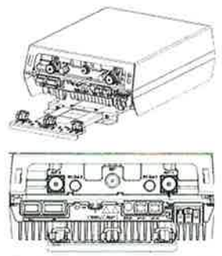
| RRH 2x40W 730 MHz COVP | |
|---|---|
| DEPTH | 5.83 IN |
| LENGTH | 18.85 IN |
| WIDTH | 16.08 IN |
| NET WEIGHT | 19.0 LBS |

**COVP SPECIFICATIONS** | SCALE N.T.S. | 1



| RRH 2x40W 730 MHz FRIG | |
|---|---|
| DEPTH | 5.98 IN |
| LENGTH | 18.88 IN |
| WIDTH | 15.24 IN |
| NET WEIGHT | 57.32 LBS |

**FRIG SPECIFICATIONS** | SCALE N.T.S. | 2



| RRH 2x40W 730 MHz FRLB | |
|---|---|
| DEPTH | 5.90 IN |
| LENGTH | 15.75 IN |
| WIDTH | 15.75 IN |
| NET WEIGHT | 59.52 LBS |

**FRLB SPECIFICATIONS** | SCALE N.T.S. | 3

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SITE INFORMATION

**SOUTHDOWN LYONS**
**DN03035D**
**5134 UTE HIGHWAY**
**LYONS, CO**
**80540**

**EQUIPMENT UPGRADE**

SHEET TITLE:

# ANTENNA PLANS

SHEET NUMBER:

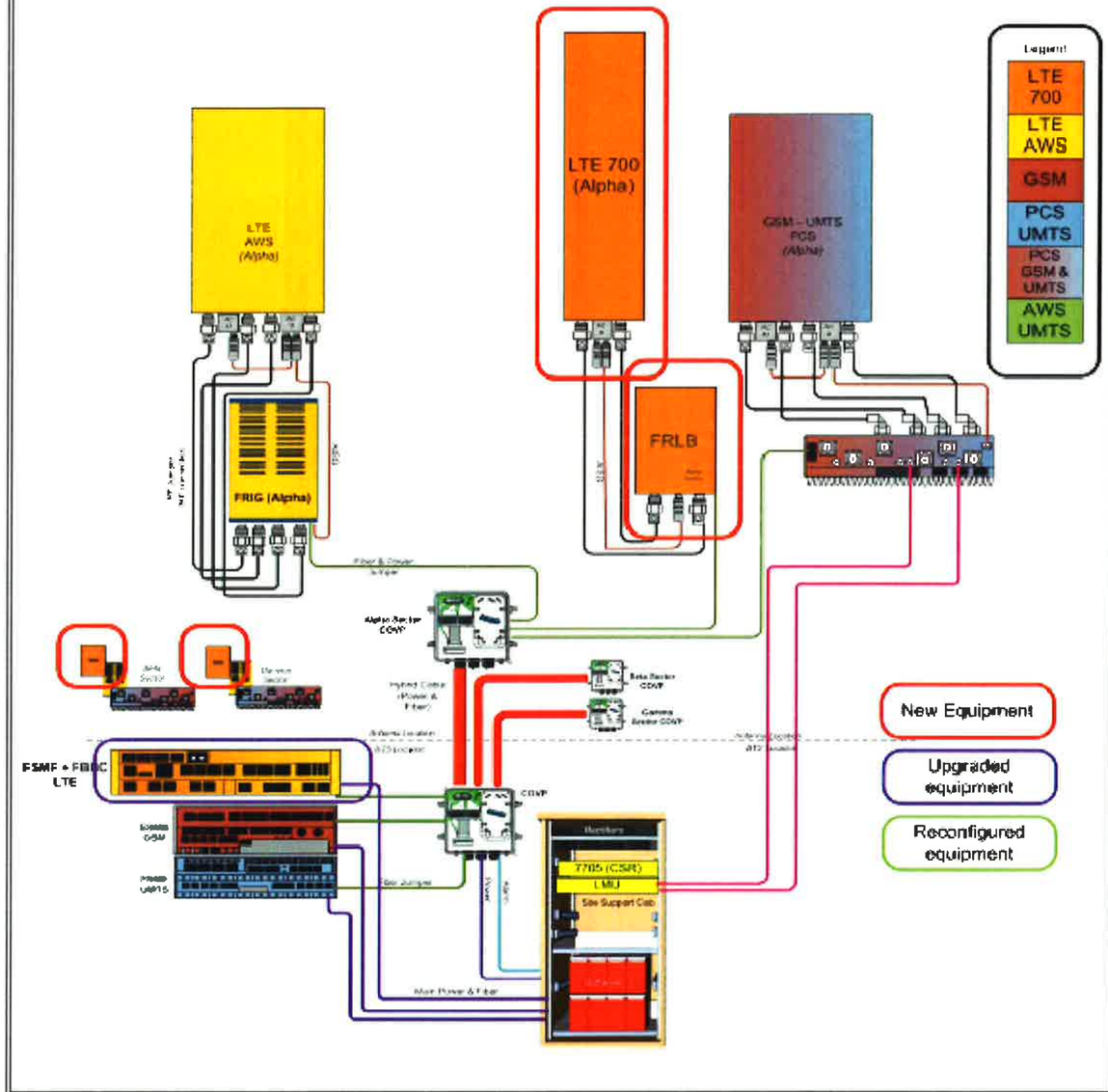
# A-5

Site: DN03035D - Configuration Drawing 1

# Drawing (1) Comments:

## Configuration 702A RUN_No U2100
## (RRU FRLB on its own antenna)

Rooftop-RRU for Contiguous PCS Spectrum Markets
(Antenna & cabling of one sector shown for clarity)






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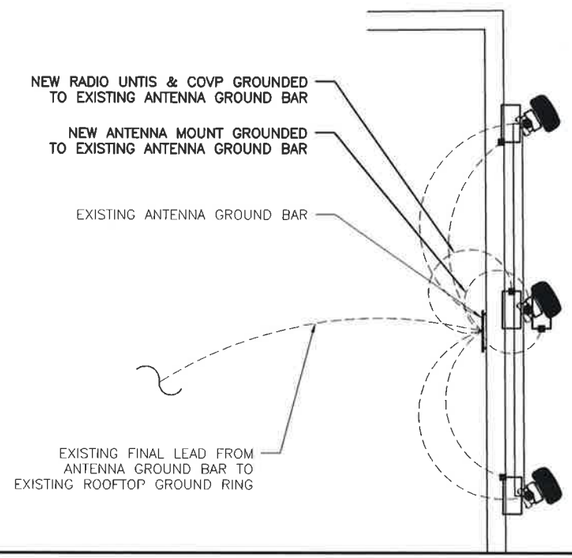
SHEET TITLE:
RF CONFIGURATION

SHEET NUMBER:
RF-1

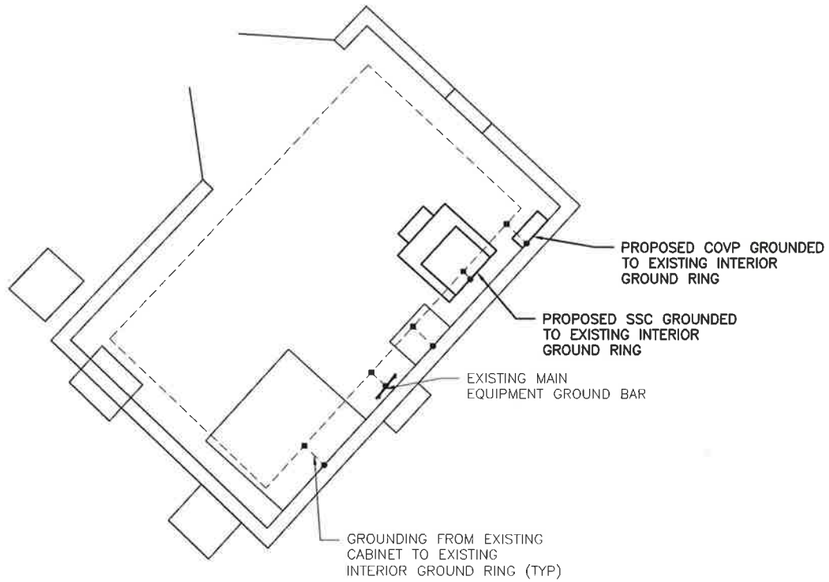
## ELECTRICAL GROUNDING SPECIFICATIONS:

1. GROUNDING SHALL COMPLY WITH ARTICLE 250 OF THE NATIONAL ELECTRICAL CODE CURRENTLY IN EFFECT FOR THE AUTHORITY HAVING JURISDICTION.
2. ALL GROUNDING DEVICES SHALL BE U.L LISTED FOR THEIR INTENDED USE.
3. GROUND WIRES SHALL BE TINNED #2 AWG BARE SOLID COPPER UNLESS OTHERWISE NOTED.
4. CONNECTIONS OF ALL GROUND WIRES TO THE GROUND RING SHALL BE EXOTHERMIC (CAD–WELDED), UNLESS OTHERWISE NOTED AND SHALL BE INSTALLED IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS AND AT&T WIRELESS BROADBAND STANDARDS.
5. GROUNDING CONDUCTORS SHALL BE ROUTED ALONG THE SHORTEST AND STRAIGHTEST PATH POSSIBLE WHEN REQUIRED. GROUND LEADS SHALL BE BENT TO A MINIMUM OF 8' RADIUS.
6. WHERE GROUND WIRES ARE ROUTED FROM ANY CONNECTION ABOVE GRADE TO THE GROUND RING, INSTALL WIRE IN 314' HEAVY WALL LIQUID TIGHT FLEXIBLE CONDUIT FROM CONNECTION POINT TO 5' BELOW GRADE AND SEAL THE TOP WITH SILICONE SEALANT.
7. ALL GROUND BARS SHALL BE TINNED, 114' COPPER, SECTOR BARS 2", COLLECTOR AND MGB BARS 4", OF SUFFICIENT LENGTH TO ACCOMMODATE ALL REQUIRED CONNECTIONS WITHOUT DOUBLING LIGS, AND EACH INSTALLED WITH ISOLATORS. WHEN CONNECTING GROUND BARS (WITHIN 10 FEET OF GRADE) DIRECTLY TO THE GROUND RING, 2 EA. #2 SOLID DOWNLEADS SHALL BE CAD–WELDED TO THE GROUNDING, 1 AT EACH OPPOSITE BOTTOM CORNER, AND EACH SHALL RUN IN 3/4" HEAVY WALL LIQUID TIGHT FLEXIBLE CONDUIT FROM GROUND BAR DOWN TO THE GROUND RING, WHEN CONNECTING SECTOR GROUND BARS, DAISY–CHAIN THE GROUND BARS AND RUN 1 EA. #2 AWG STRANDED COPPER WIRE WITH THWN INSULATION FROM THE MIDDLE GROUND BAR TO THE GROUND RING AND CAD–WELD TO THE RING.
8. WHEN ATTACHING STRANDED GROUND LEADS TO THE GROUND BARS, 2 HOLE COMPRESSION LUGS SHALL BE USED, PROTECT WITH WEATHERPROOF HEAT SHRINK, AND WITH A THIN COAT OF "KOP'R SHIELD' OR EQUIVALENT PROPERLY APPLIED AND ATTACHED ONLY WITH STAINLESS STEEL HARDWARE.
9. WHEN GROUNDING EQUIPMENT ENCLOSURES, PANELS , FRAMES, AND OTHER METAL APPARATUS, A #6 AV/G STRANDED COPPER WIRE WITH THWN INSULATION SHALL BE ATTACHED UTILIZING A 2 HOLE COMPRESSION TYPE LUG, PROTECTED WITH WEATHERPROOF HEAT A CLEAN AND CORROSION FREE METALLIC SURFACE UTILIZING STAINLESS STEEL SELF–TAPPING SCREWS AS NOTED IN NOTE 10 BELOW.
10. PREPARE ALL BONDING SURFACES FOR GROUND CONNECTIONS BY REMOVING ANY AND ALL PAINT AND CORROSION TO SHINY METAL. CAD–WELDED CONNECTIONS TO NON–GOPPER SURFACES, APPLY ONE COAT OF ANY ANTI–OXIDIZING PAINT, "COLD GALV" OR EQUIVALENT.
11. GROUND RODS SHALL BE COPPER–CLAD STEEL 5/8"x10', SPACED NO LESS THAN 10' ON CENTER.
12. ALL GROUND SYSTEM CONDUCTORS AND CONDUITS SHALL BE SECURED UTILIZING ONLY NONMETALLIC, NON–CONDUCTIVE, UV RATED CLAMPS, BRACKET, AND OR SUPPORTS.
13. WHEN REQUIRED, THE CONTRACTOR SHALL ENGAGE THE SERVICES OF AN INDEPENDENT TESTING FIRM TO VERIFY, UTILIZING A MEGGER TEST, THAT THE RESISTANCE TO EARTH OF THE NEW GROUND SYSTEM IS EQUAL TO OR LESS THAN 5 (OHMS). A COPY OF THE COMPLETE TESTING REPORT SHALL BE PROVIDED TO THE AT&T REPRESENTATIVE.
14. ALL MATERIALS AND HARDWARE SHALL BE INSTALLED IN A WORKMAN–LIKE MANNER IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS, AND DEFINED IN NFPA–70.

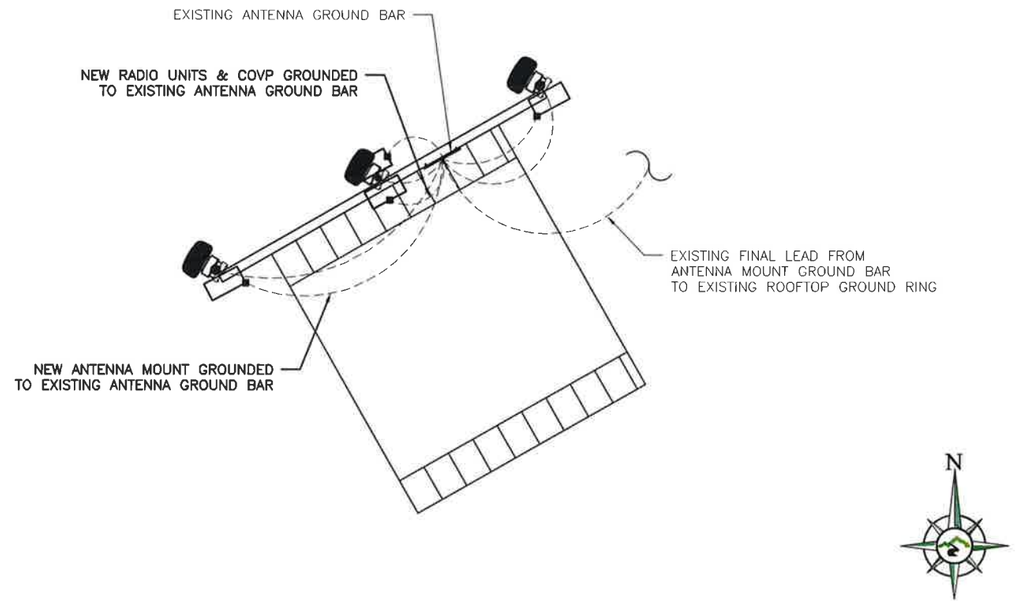
GROUNDING NOTES | 3




ANTENNA GROUNDING (BETA SECTOR) | SCALE N.T.S. | 4




EQUIPMENT GROUNDING | SCALE N.T.S. | 1




ANTENNA GROUNDING (ALPHA & GAMMA SECTORS) | SCALE N.T.S. | 2

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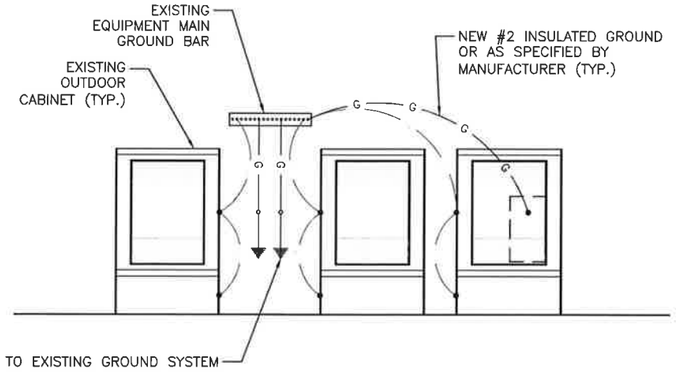
SITE INFORMATION

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**80540**

**EQUIPMENT UPGRADE**

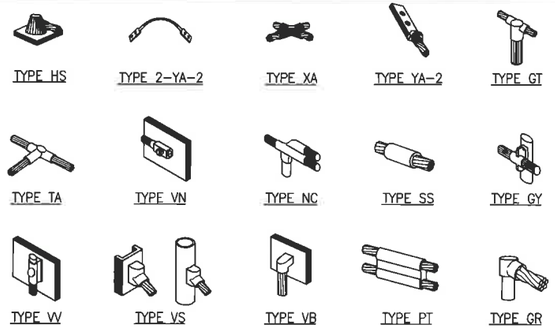
SHEET TITLE:
GROUNDING DETAILS

SHEET NUMBER:
G-1




## GROUNDING SCHEMATIC

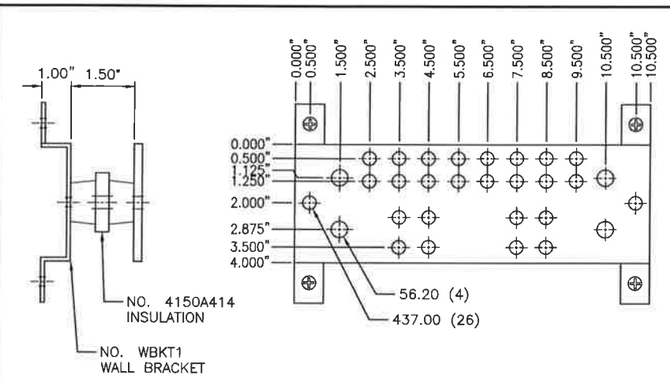
SCALE N.T.S. 7




NOTE:
ERICO EXOTHERMIC "MOLD TYPES" SHOWN HERE ARE EXAMPLES. CONSULT WITH PROJECT MANAGER FOR SPECIFIC MOLDS TO BE USED FOR THIS PROJECT.

## EXOTHERMIC WELDING

SCALE N.T.S. 4

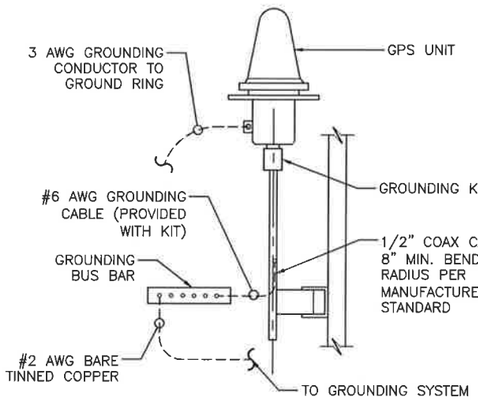



## 12" & 18" GROUND BAR

SCALE N.T.S. 1

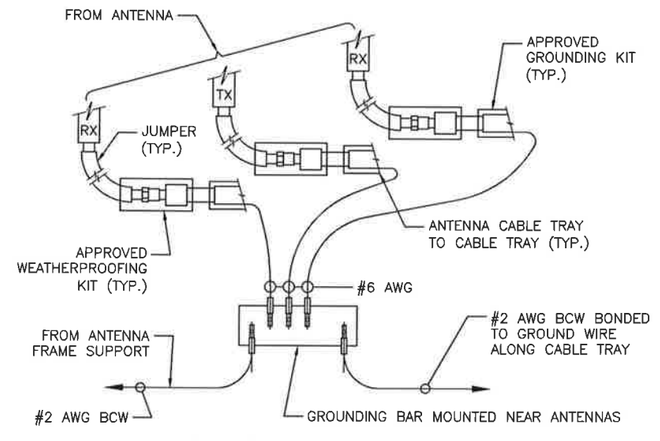
## NOT USED

SCALE N.T.S. 8




## GPS ANTENNA MOUNTING

SCALE N.T.S. 5



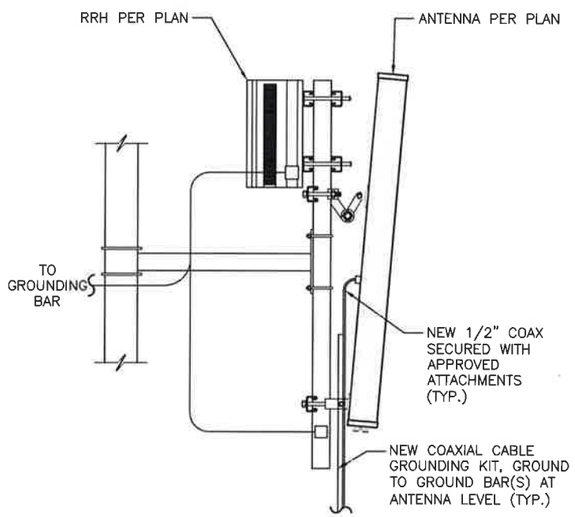

NOTE: DO NOT INSTALL CABLE GROUND KIT AT A BEND

## ANTENNA GROUND KIT

SCALE N.T.S. 2

## NOT USED

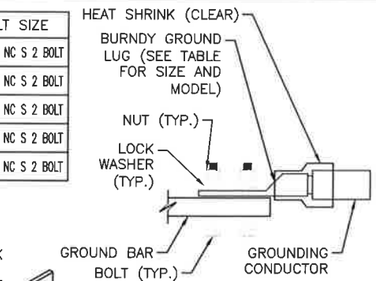
SCALE N.T.S. 9




## ANTENNA GROUNDING

SCALE N.T.S. 6

| WIRE SIZE | BURNDY LUG | BOLT SIZE |
|---|---|---|
| #6 AWG GREEN INSULATED | YA6C-2TC38 | 3/8"-16 NC S 2 BOLT |
| #2 AWG SOLID TINNED | YA3C-2TC38 | 3/8"-16 NC S 2 BOLT |
| #2 AWG STRANDED | YA2C-2TC38 | 3/8"-16 NC S 2 BOLT |
| #2/0 AWG STRANDED | YA26-2TC38 | 3/8"-16 NC S 2 BOLT |
| #4/0 AWG STRANDED | YA28-2N | 3/8"-16 NC S 2 BOLT |




NOTES:

1. ALL HARDWARE BOLT, NUTS, LOCK WASHERS SHALL BE STAINLESS STEEL. ALL HARDWARE ARE TO BE AS FOLLOWS: BOLT, FLAT WASHER, GROUND BAR, GROUND LUG, FLAT WASHER AND NUT.
2. COPPER SHIELD, ANTIOX, CR NO-OX OR EQUIVALENT SHALL BE PLACED WHERE ALL DISSIMILAR METALS CONNECT.
3. ALL LUGS TKO BE INSTALLED PER MANUFACTURER'S SPECIFICATIONS.

## MECHNICAL LUG CONNECTION

SCALE N.T.S. 3

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**5134 UTE HIGHWAY**
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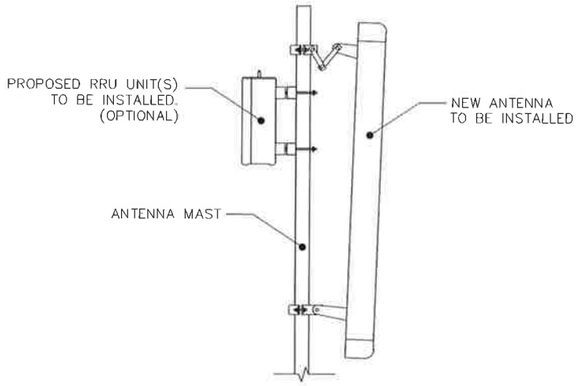
**EQUIPMENT UPGRADE**

SHEET TITLE:

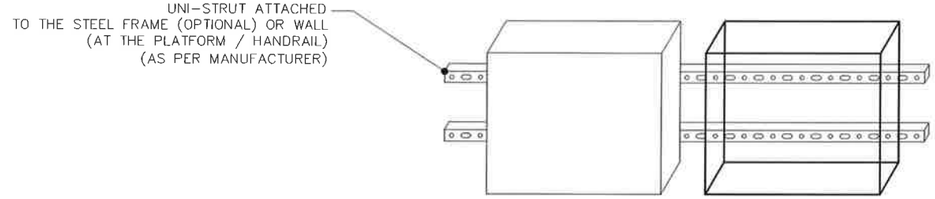
GROUNDING DETAILS

SHEET NUMBER:

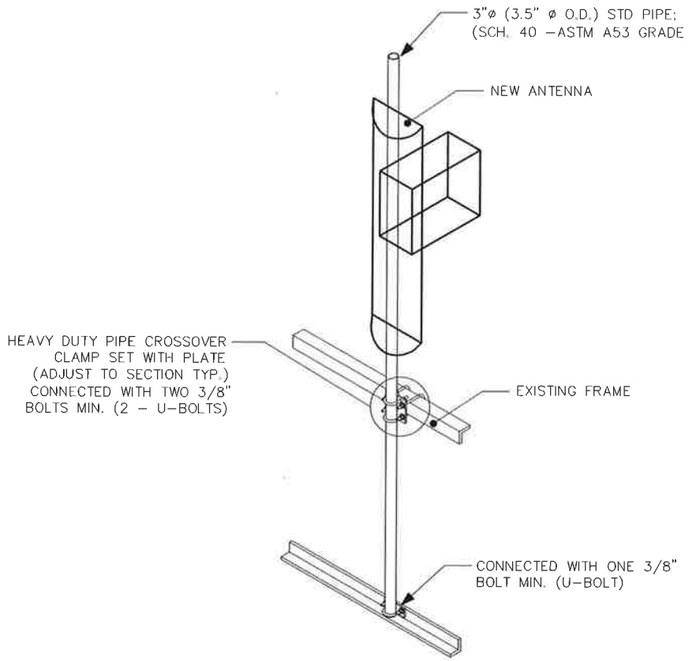
G-2




**TYPICAL MODULE INSTALLATION (OPTION 2)** | SCALE N.T.S. | 3



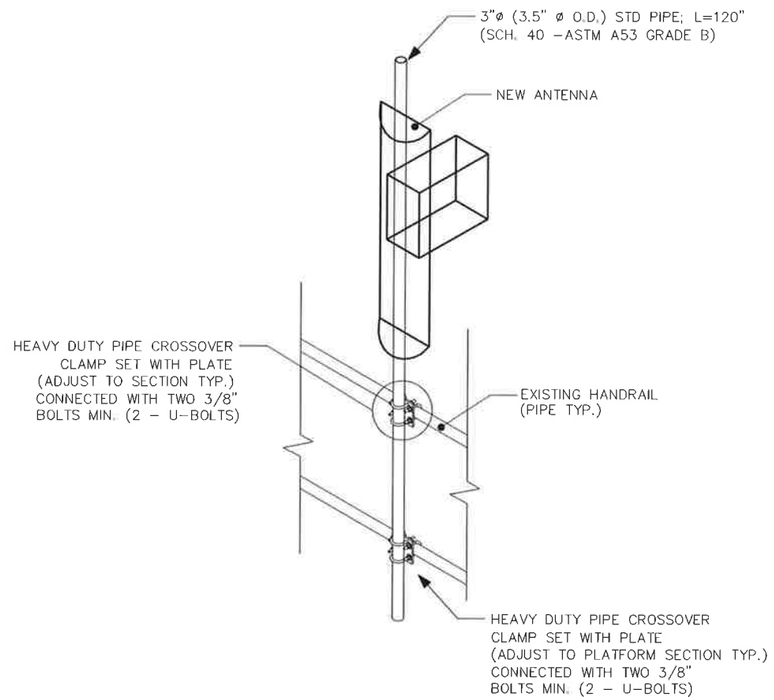

**TYPICAL MODULE INSTALLATION (OPTION 1)** | SCALE N.T.S. | 1




NOTES:

1. ALL MATERIALS SHALL BE HOT DIP GALVANIZED STEEL A36 (TYP.)
2. USE GALVANIZED HEAVY HEX NUTS, LOCK AND FLAT WASHERS AS REQUIRED.
3. THE CONTRACTOR SHALL VERIFY THAT ALL EXISTING MEMBERS AND PARTS ARE IN GOOD CONDITION AND STRUCTURALLY SOUND PRIOR INSTALLATION.
4. THE CONTRACTOR SHALL VERIFY THAT ALL DIMENSIONS ARE ADEQUATE PRIOR TO INSTALLATION.

**PROPOSED CONDITIONS (ALPHA AND GAMMA)** | SCALE N.T.S. | 4




NOTES:

1. ALL MATERIALS SHALL BE HOT DIP GALVANIZED STEEL A36 (TYP.)
2. USE GALVANIZED HEAVY HEX NUTS, LOCK AND FLAT WASHERS AS REQUIRED.
3. THE CONTRACTOR SHALL VERIFY THAT ALL EXISTING MEMBERS AND PARTS ARE IN GOOD CONDITION AND STRUCTURALLY SOUND PRIOR INSTALLATION.
4. THE CONTRACTOR SHALL VERIFY THAT ALL DIMENSIONS ARE ADEQUATE PRIOR TO INSTALLATION.

**PROPOSED CONDITIONS (BETA)** | SCALE N.T.S. | 2

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**EQUIPMENT UPGRADE**

SHEET TITLE:

STRUCTURAL DETAILS

SHEET NUMBER:

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